BRSM Mid Project Report

Team - P < 1, Github -
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1 Introduction

This project investigates how sentence memorability is influenced by the memorability of constituent nouns and syntactic voice. Using a continuous recognition memory paradigm, participants were presented with sentences systematically varied across four noun-memorability conditions (HH, HL, LH, LL) and two syntactic voices (Active and Passive). Statistical analyses are used to evaluate hypotheses regarding the effect of noun composition and sentence voice on recognition performance. The study aims to demonstrate how quantitative analysis of experimental data can reveal patterns in sentence recall and provide insights into the cognitive processes underlying memory for language.

2 Data Overview and Description

2.1 Data Source

The dataset was collected from 114 students enrolled in the BRSM course at IIIT Hyderabad. Each participant completed an online recognition memory experiment, with responses recorded in anonymized log files. The experiment was designed and administered by researchers from the CogSci Lab@IIITH.

The logs contain detailed trial-level information including stimulus identifiers, event types, response accuracy, and reaction times. These records form the basis for the preprocessing, exploratory analysis, and statistical testing performed in this study.

2.2 Dataset Description

The dataset consists of 114 anonymized log files, each representing a participant's responses in the sentence memorability experiment. Each log file contains detailed trial-level records, including event types, stimulus information, response accuracy, reaction times, and other relevant metadata. The columns are described below:

3 Domain Background

Human memory has long been a central topic in cognitive psychology. Researchers aim to understand how individuals encode information, retain it over time, and retrieve it when needed. Sentence memory is particularly interesting because it involves both the storage of meaning and the reconstruction of linguistic structure.

Early theoretical models proposed that sentences are stored as a core semantic representation, sometimes referred to as a "kernel," along with additional mental tags that represent grammatical features such as passive or negative forms. Under this view, memory for sentence structure was considered an "all-or-none" process: a person either remembered the syntactic tag or failed to recall it entirely.

However, research by Bregman and Strasberg (1968) challenged this interpretation. Their experiments showed that even when participants failed to remember the exact syntactic form of a sentence, their responses still contained meaningful structural information. This suggested that sentence memorability is not based solely on discrete grammatical tags, but rather on a reconstructed semantic representation derived from partial memory traces.

Column	Description
participant_ID	Anonymized numeric participant identifier
Timestamp	Event time in milliseconds
Event	Type of event (e.g., Sentence shown, IR pressed, WR pressed, Missed, etc.)
Stimulus	Sentence ID encoding condition, number, and voice (e.g., HH_130_A)
isTarget	True if the sentence is a target (will appear twice)
isValidation	True if this is a validation/attention-check sentence
isRepeat	True if this is the second (recognition) presentation
Button	Response key (e.g., Spacebar, Yes/No, GapTime_XXXX, Rest)
Accuracy IR	1 = correct IR response, 0 = incorrect
Accuracy WR	1 = correct WR response, 0 = incorrect
Reaction_time_IR	Milliseconds from sentence onset to Spacebar press
Reaction_time_WR	Milliseconds from WR prompt to Yes/No click
CR_IR / CR_WR	Cumulative recall columns (reserved, currently empty)

Table 1: Dataset column descriptions

Further evidence was provided by Begg (1971), who demonstrated that memory for sentence meaning and memory for exact wording are largely independent processes. His findings support the idea that concrete sentences are often stored as mental imagery or conceptual representations, while the specific wording of the sentence is reconstructed during retrieval.

James, Thompson, and Baldwin (1973) further emphasized that this reconstruction process is guided by natural linguistic biases. Their work showed that recall is influenced by noun saliency, where individuals tend to reconstruct sentences around a vivid or highly memorable noun that acts as the central "theme" of the sentence. Because of the cognitive preference for active voice structures, sentences are often recalled more easily when the salient noun appears in the subject position.

Together, these findings suggest that sentence memorability emerges from the interaction between semantic representation, linguistic structure, and cognitive reconstruction processes. The present study builds on this theoretical background by experimentally examining how noun memorability and sentence voice influence recognition performance in a controlled sentence memory task.

4 Methods

4.1 Exploratory Data Analysis

The dataset initially consisted of **81,329 recorded event rows collected from 114 participants**.After removing practice trials, **76,060 rows remained for analysis**. After applying block-level validation filtering, the final dataset contained **73,503 usable rows**.

This distribution ensured that each experimental condition was sufficiently represented in the dataset.

Condition	Event Count
HF	29,733
HH	8,589
HVL / HL	7,258
LVH	5,130
LVL	5,077

Table 2: Distribution of stimulus prefixes in the dataset

The exploratory analysis focused on three main aspects:

Dataset Structure: The log files contained multiple event types including sentence presentation, recognition responses, validation checks, and reaction time measurements. Each stimulus identifier encoded the experimental condition, sentence ID, and syntactic voice.

Stimulus Distribution: The dataset contained several stimulus categories corresponding to noun memorability combinations. The event distribution across prefixes was as follows:

Initial Performance Patterns:

Histograms of participant-level memorability scores showed that most participants had corrected scores clustered around **0.7**, indicating generally high recognition accuracy. Reaction time distributions for correct recognition responses were also inspected, showing mean response times between **1.6 and 1.8 seconds** across conditions. These exploratory steps helped verify data integrity and provided initial insights into recognition performance before formal hypothesis testing.

4.2 Data Acquisition and Preprocessing

A total of **114 participants** were recruited from the BRSM course and were initially provided with a practice set to familiarize themselves with the experimental task. Following this, participants were presented with **48 target sentences across three blocks**. These stimuli were systematically drawn from a pool of **400 short sentences** (each sentence ≤ 8 words), categorized by noun-memorability (**HH, HL, LH, LL**) and syntactic voice (**Active, Passive**).

Sentences were preprocessed through automated fluency checks, semantic-coherence filters, and manual screening to ensure strict grammaticality and plausibility. To maintain data integrity, blocks that failed the following validation formula were excluded from the final analysis:

$$CorrectIRs > \left(\frac{WrongIRs}{2}\right) + MissedIRs \tag{1}$$

This exclusion criterion ensured high-quality data for calculating corrected memorability scores. As a result, **10 blocks were excluded** and **1 participant was removed entirely** from the analysis.

4.3 Experimental Setting

Participants completed a **continuous recognition memory experiment** conducted through a web-based interface. Each participant first completed a practice session to familiarize themselves with the task, followed by the main experiment consisting of **three experimental blocks**.

The stimulus pool contained **400 short sentences which are present in both active and passive form($\leq$ 8 words)** categorized according to noun memorability which were - HH, HL, LH and LL (H - high and L - Low). The experiment flow was as follows -

1. A sentence was displayed on the screen.
2. Participants pressed the **spacebar** if they recognized the sentence from a previous presentation.
3. After pressing the recognition key, they indicated whether the sentence was **exactly the same or structurally different** from the earlier version (key **a** and key **d** were used).

Each participant encountered **48 target sentences distributed across three blocks**, with short breaks allowed between blocks to reduce fatigue.

4.4 Corrected Memorability Score

Recognition performance was quantified using a **corrected memorability score**, which adjusts for response bias. Two measures were computed for each participant and condition.

$$HitRate = \frac{Correct\ IR\ Hits}{Total\ Repeat\ Targets} \tag{2}$$

Hit Rate represents the proportion of repeated sentences that participants correctly recognized.

$$FalseAlarmRate = \frac{Incorrect\ IR\ Presses\ on\ Nonrepeat\ Trials}{Total\ Nonrepeat\ Trials} \tag{3}$$

False Alarm Rate represents cases where participants incorrectly indicated that a sentence had been seen before.

$$CorrectedScore = HitRate - FalseAlarmRate \tag{4}$$

The Corrected adjustment ensures that the memorability measure reflects **true recognition ability rather than a participant's tendency to press the recognition key frequently**. Corrected memorability scores were computed for each participant and condition and were used as the primary dependent variable for both descriptive analysis and hypothesis testing.

5 Descriptive Analysis

Descriptive analysis was conducted to summarize the overall structure and performance patterns of the dataset before performing inferential statistical tests.

The final dataset used for analysis contained **73,503 event rows from 113 participants**, after removing practice trials and excluding invalid experimental blocks.

5.1 Recognition Performance

Recognition performance was evaluated using **repeat hit rates and miss rates** aggregated at the participant level.

Condition	Mean Hit Rate
HH	0.846
HL	0.847
LH	0.837
LL	0.788

Table 3: Mean repeat hit rates across conditions

Condition	Mean Miss Rate
HH	0.154
HL	0.153
LH	0.163
LL	0.212

Table 4: Mean repeat miss rates across conditions

These results indicate that sentences in the **LL condition produced the lowest recognition accuracy and the highest miss rates**, suggesting that sentences composed entirely of low-memorability nouns are more difficult for participants to remember.

5.2 Corrected Memorability Scores

After accounting for false alarms, the mean corrected memorability scores across conditions were:

Condition	Mean Corrected Score
HH	0.724
HL	0.725
LH	0.716
LL	0.666

Table 5: Mean corrected memorability scores

The results show that the **LL condition has a noticeably lower corrected score compared to the other three conditions**, indicating reduced memorability when both nouns in the sentence have low memorability.

5.3 Reaction Time Statistics

Reaction times were analyzed for correct recognition responses.

Condition	Mean Reaction Time (ms)
HH	1725
HL	1712
LH	1661
LL	1761

Table 6: Mean reaction times for correct recognition responses

Reaction times across conditions fall within a relatively narrow range of **1.6–1.8 seconds**, suggesting that differences between conditions primarily affect **recognition accuracy rather than response speed**.

5.4 Visualization of Data

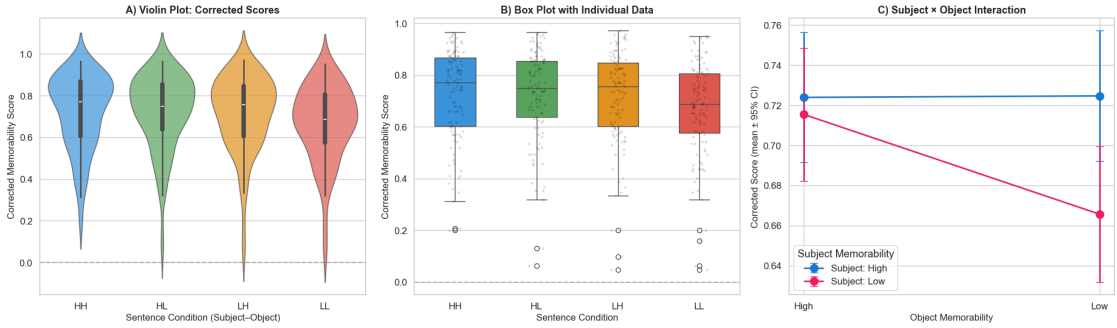

Figure 1: Corrected Memorability Scores Across Conditions

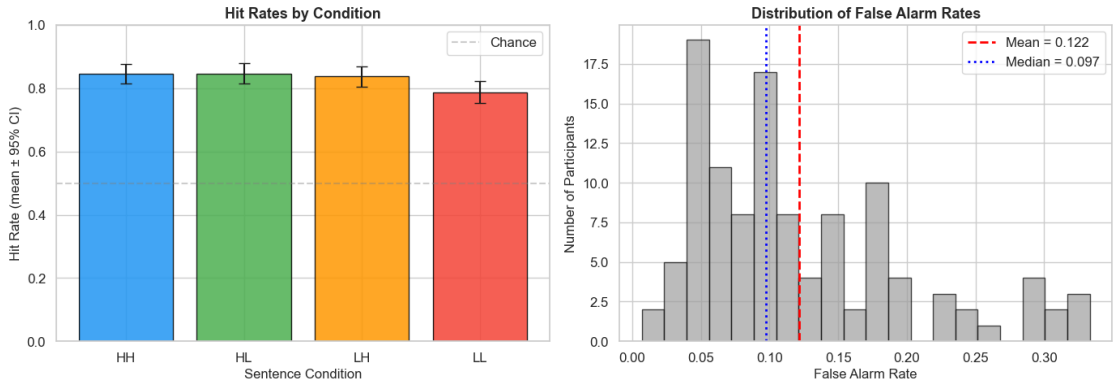

Figure 2: Distribution of Hit Rate and False Alarm Rate

To further examine these patterns, several visualizations were generated:

- Histograms of participant-level corrected scores showing most participants clustered around a score of approximately **0.7**.
- Boxplots of corrected memorability scores across conditions highlighting the comparatively lower performance of the **LL condition**.
- Line plots of performance across experimental blocks indicating a slight decline in performance from Block 1 to Block 3.
- Reaction-time distribution plots confirming that response speed remained relatively stable across conditions.

These descriptive statistics provide an initial overview of the dataset and reveal the main performance patterns that are examined more formally through hypothesis testing in the following section.

6 Hypothesis Testing

6.1 Hypotheses

H1 (Noun Memorability Effect): Sentences with higher-memorability nouns (HH) will show higher corrected memorability scores than those with lower-memorability nouns (LL), with mixed conditions (HL, LH) falling in between.

H0$_1$: The four conditions (HH, HL, LH, LL) produce equal corrected memorability scores.

H2 (Voice Effect): Sentence voice (Active vs. Passive) affects recognition memorability.
H0$_2$: There is no difference in memorability between active and passive voice sentences.

6.2 Normality Check (Shapiro-Wilk Test)

Before selecting inferential tests, we checked whether corrected memorability scores follow a normal distribution using the Shapiro-Wilk test. If $p < 0.05$, the data significantly departs from normality.

Condition	W Statistic	p-value	Normal?
HH	0.924	< 0.001	No
HL	0.914	< 0.001	No
LH	0.919	< 0.001	No
LL	0.940	< 0.001	No

Table 7: Shapiro-Wilk normality test results for each condition

All conditions violate normality ($p < 0.001$). Q-Q plots (Figure 3) confirm left-skewed distributions with a ceiling effect. We therefore use **non-parametric** tests throughout: **Kruskal-Wallis** H (4-group comparison), **Dunn's post-hoc** with Bonferroni correction (pairwise follow-up), and **Mann-Whitney** U (2-group comparison).

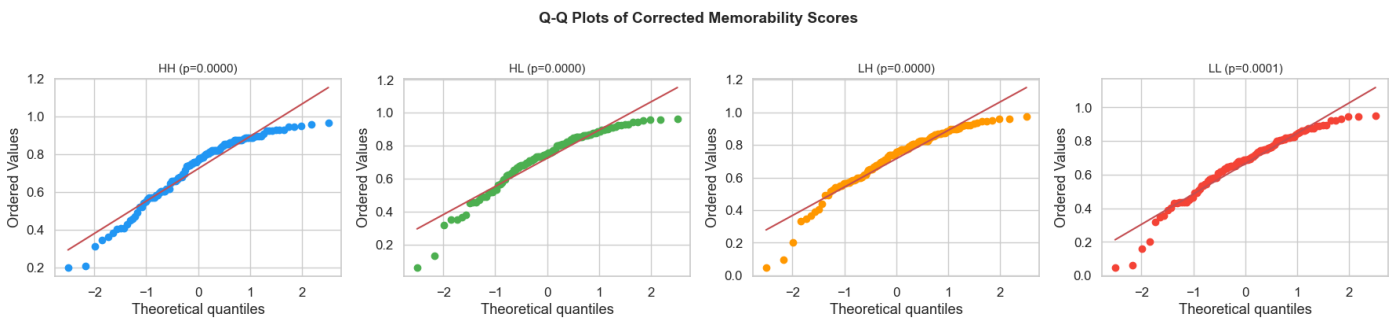

Figure 3: Q-Q plots of corrected memorability scores. Deviations from the diagonal indicate departures from normality; all four conditions show left-skewed distributions.

6.3 Testing H1: Noun Memorability Effect (Kruskal-Wallis)

The Kruskal-Wallis H test is the non-parametric equivalent of one-way ANOVA. It pools all observations, converts them to ranks, and tests whether mean ranks differ across groups. Under the null hypothesis, H follows a χ^2 distribution with $k - 1$ degrees of freedom.

Results

Measure	Value
H statistic	10.0014
Degrees of freedom	3
p-value	0.0186
Effect size η^2_H	0.0156
N	452

Table 8: Kruskal-Wallis test results for the noun memorability effect

Since $p = 0.019 < 0.05$, we **reject** $H0_1$. Corrected memorability scores differ significantly across conditions. The effect size ($\eta^2_H = 0.016$) indicates noun memorability explains about 1.6% of the variance — a small but real effect.

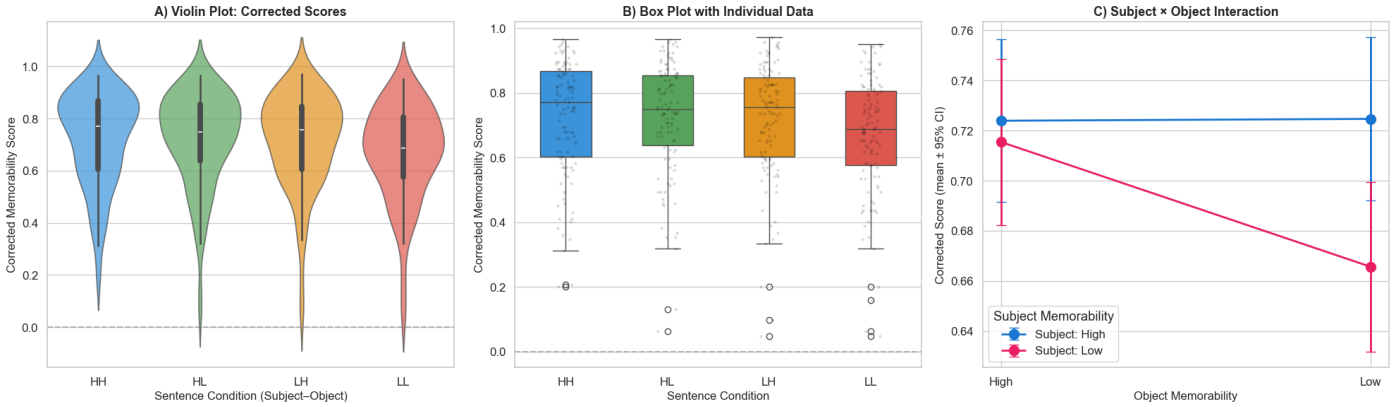

Figure 4: Corrected memorability scores across conditions. (A) Violin plot. (B) Box plot with individual data points. (C) Subject × Object memorability interaction (mean ± 95% CI). LL is visibly lower in all panels.

6.4 Post-Hoc Pairwise Comparisons (Dunn's Test)

The Kruskal-Wallis test indicates *at least one* condition differs but not *which*. We used Dunn's test for all $\binom{4}{2} = 6$ pairwise comparisons. To control the family-wise error rate (which inflates to ∼26.5% over 6 tests), each raw p-value is multiplied by 6 (**Bonferroni correction**).

Comparison	Bonferroni p	Holm p	Significant?
@HH vs HL	1.000	1.000	No
HH vs LH	1.000	1.000	No
HH vs LL	**0.041**	**0.041**	**Yes**
HL vs LH	1.000	1.000	No
HL vs LL	**0.043**	**0.041**	**Yes**
LH vs LL	0.148	0.098	No (trend)

Table 9: Dunn's post-hoc pairwise comparisons with Bonferroni and Holm corrections

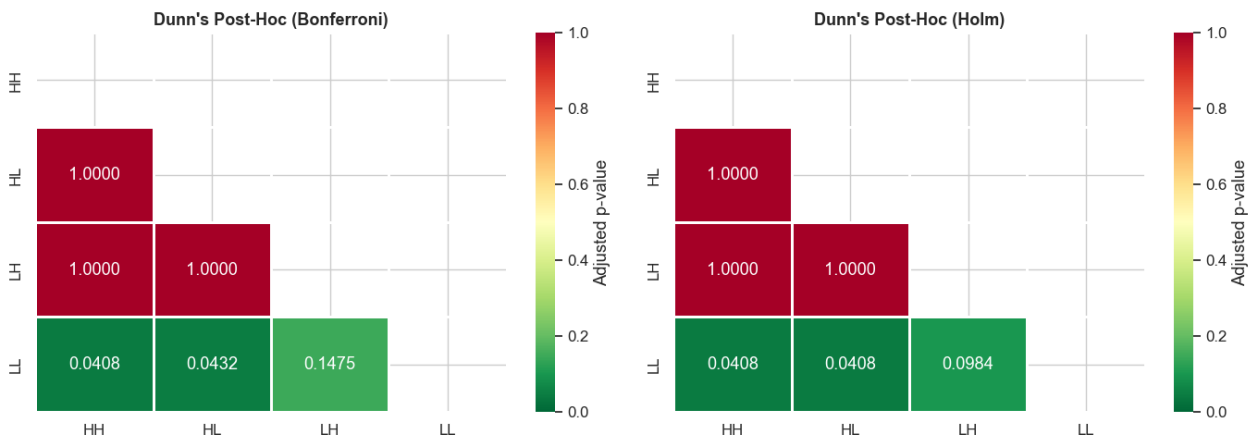

Figure 5: Heatmaps of Dunn's post-hoc adjusted p-values. Green = significant ($p < 0.05$); red = non-significant.

6.4.1 Interpretation of H1

- **HH vs LL ($p = 0.041$):** Significant — directly supports H1.

- **HL vs LL ($p = 0.043$):** Significant — a high-memorability subject noun alone suffices to outperform LL.
- **LH vs LL ($p = 0.148$):** Not significant, though a trend exists.
- **HH vs HL, HH vs LH, HL vs LH ($p = 1.0$):** The top three conditions are statistically indistinguishable.

Observed ordering of mean corrected scores:

$$\text{HL } (0.725) \approx \text{HH } (0.724) > \text{LH } (0.716) > \text{LL } (0.666)$$

Conclusion for H1: Partially supported. There is a clear divide between conditions containing at least one high-memorability noun (HH, HL, LH) and the all-low condition (LL). The effect follows a **threshold pattern** rather than a graded one — one high-memorability noun provides a significant benefit, but a second does not add further improvement. $H0_1$ is **rejected**.

6.5 Testing H2: Voice Effect (Mann-Whitney U)

The Mann-Whitney U test is the non-parametric equivalent of the independent t-test. It ranks all observations from both groups and tests whether the rank distributions differ. The rank-biserial correlation r serves as the effect size.

Measure	Value
Active voice mean corrected score	0.702
Passive voice mean corrected score	0.713
U statistic	6055.50
p-value	0.504
Rank-biserial r	0.052

Table 10: Mann-Whitney U test results for the voice effect

Since $p = 0.504 \gg 0.05$, we **fail to reject** $H0_2$. Voice has no significant effect on memorability, with a negligible effect size ($r = 0.052$). H2 is **not supported**.

6.5.1 Condition × Voice Interaction (Exploratory)

A Kruskal-Wallis test across all 8 groups (4 conditions × 2 voices) yielded $H(7) = 14.03$, $p = 0.051$ — marginally non-significant. The parallel lines in Figure 6 confirm no meaningful interaction.

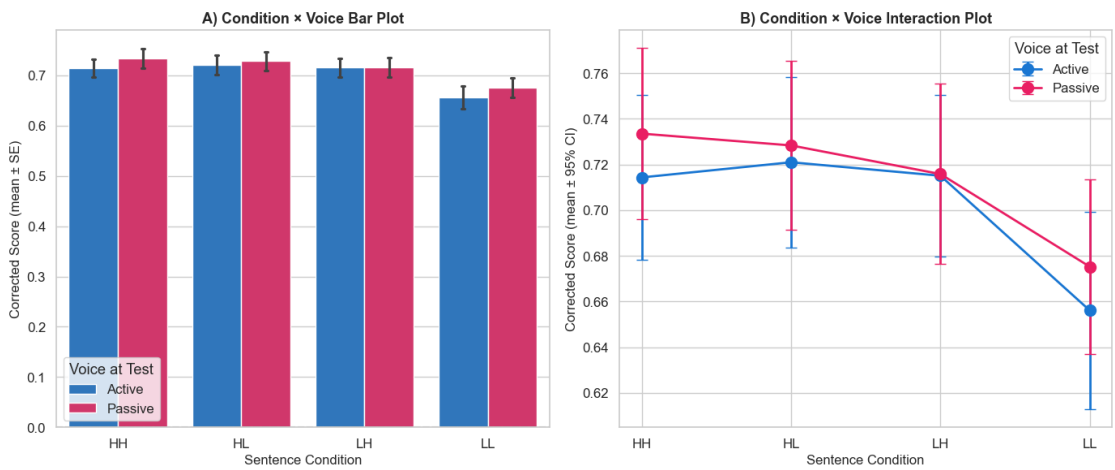

Figure 6: Condition × Voice interaction. (Left) Grouped bar plot. (Right) Interaction plot showing parallel lines, indicating no interaction.

7 Further Works

While the current study provides initial insights into the relationship between noun memorability and sentence recognition, several extensions can be explored in the final report to strengthen the analysis and interpretation.

1. Investigate additional hypotheses involving **reaction time and recognition accuracy across experimental blocks** to analyze possible fatigue or learning effects during the experiment.
2. Perform deeper statistical analysis using **non-parametric methods and alternative robustness checks** to validate the consistency of the observed patterns.
3. Explore **signal detection based metrics** such as d' (d-prime) and response bias measures to better understand recognition sensitivity and participant decision strategies.
4. Review additional literature in cognitive psychology and sentence memorability research to identify **alternative evaluation metrics and modeling approaches**.
5. Apply more advanced statistical models such as **mixed-effects models or item-level analyses** to simultaneously account for participant variability and sentence-level differences.
6. Examine potential **interaction effects between noun memorability and syntactic voice**, as well as other linguistic properties such as sentence length or word frequency.

8 Conclusion

8.1 Key Takeaways

This study examined how sentence memorability is affected by noun memorability and syntactic voice using a continuous recognition memory experiment with 114 participants.

The results show that noun memorability has a significant effect on recognition performance. The Kruskal–Wallis test indicated differences across the four conditions ($H = 10.0014$, $p = 0.0186$), with post-hoc comparisons showing that the LL condition had significantly lower memorability than HH and HL.

The corrected memorability scores followed the pattern:

$$HL \approx HH > LH > LL$$

This suggests that having at least one high-memorability noun improves sentence recognition.

In contrast, the Mann–Whitney U test found no significant effect of sentence voice ($p = 0.504$). Overall, the findings indicate that noun memorability plays a larger role in sentence recall than syntactic voice.

8.2 Limitations

Despite these findings, several limitations should be noted.

First, the observed effect size for noun memorability was relatively small ($\eta_H^2 = 0.016$), indicating that other linguistic or cognitive factors may also contribute to sentence memorability.

Second, the dataset was collected from a relatively homogeneous participant pool consisting primarily of students, which may limit the generalizability of the results to broader populations.

Third, the experiment focused on a limited set of sentence structures and noun combinations. Additional linguistic features such as sentence length, semantic richness, or word frequency were not explicitly analyzed.

Finally, although non-parametric statistical methods were used due to violations of normality, more advanced modeling approaches such as mixed-effects models could provide deeper insight by simultaneously accounting for participant-level and sentence-level variability.

Future work could address these limitations by expanding the dataset, incorporating additional linguistic variables, and applying more advanced statistical modeling techniques.

9 References and Contributions

9.1 References

1. Begg, I. (1971). *Recognition memory for sentence meaning and wording.* Journal of Verbal Learning and Verbal Behavior.

2. Bregman, A. S., & Strasberg, S. (1968). *Memory for the syntactic form of sentences.* Journal of Verbal Learning and Verbal Behavior.

3. James, C. T., Thompson, J. G., & Baldwin, J. M. (1973). *Reconstruction processes in sentence memory.* Journal of Verbal Learning and Verbal Behavior.

4. Snodgrass, J. G., & Corwin, J. (1988). Pragmatics of measuring recognition memory: Applications to dementia and amnesia. Journal of Experimental Psychology: General, 117(1), 34–50.

5. Gernsbacher, M. A. (1988). *Sentence participants: The advantage of first mention.* Journal of Memory and Language.

6. Course slides for BRSM-S26.

9.2 Team Contributions

1. **Arnav Deshmukh** Worked on data preprocessing and parsing of the experimental log files. Implemented parts of the exploratory data analysis and contributed to statistical testing and visualization of the results.

2. **Nidhish Jain** Contributed to the implementation of statistical analysis, including hypothesis testing using non-parametric methods such as the Kruskal–Wallis and Mann–Whitney tests. Assisted in interpreting the results and preparing figures and tables.

3. **Nilavra Ghosh** Worked on exploratory data analysis, visualization of participant performance, and preparation of report sections including descriptive analysis and interpretation of results. Assisted in organizing the dataset and compiling the final report.

   Note: All team members jointly contributed to the conceptualization of the study, experimental design, interpretation of results, and preparation of the report. The contributions listed above indicate the primary responsibilities undertaken by each member for specific analysis and implementation components.